Q4 NO CALCULATOR ALLOWED Q4

Answer QUESTION 4 PARTS A and B on this page.

PART A

$\frac{d}{dx} g(x) = \frac{d}{dx} \int_0^x f(t)\,dt$

$g'(x) = f(x)$

$g'(8) = f(8) = \boxed{1}$

fundamental theorem of calculus states that $\frac{d}{dx}\left[\int f(x)\,dx\right] = f(x)$

PART B

g has a pt of inflection at $x=-3$ because f (or g') changes from decreasing to increasing

g has a pt of inflection at $x=3$ because f (or g') changes from increasing to decreasing

g has a pt of inflection $x=6$ because f (or g') changes from decreasing to increasing

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Q4 NO CALCULATOR ALLOWED Q4

Answer QUESTION 4 PARTS C and D on this page.

PART C

$g(12) = \int_6^{12} f(t)\,dt$

$g(12) = \frac{(6)(3)}{2}$

$\boxed{g(12) = 9}$

$g(0) = \int_6^0 f(t)\,dt$

$g(0) = -\int_0^6 f(t)\,dt$

$g(0) = -\left(\frac{1}{2}\pi(3)^2\right)$

$g(0) = -\left(\frac{9}{2}\pi\right)$

$\boxed{g(0) = -\frac{9}{2}\pi}$

PART D

g is continuous on $[-6, 12]$

$\therefore$ EVT applies

$g'(x) = f(x)$

$0 = f(x)$

$x = 0 \quad x = 6$

x	$g(x)$
-6	0
0	$-\frac{9}{2}\pi$
6	0
12	9

$g(0) < g(-6) = g(6) < g(12)$

$\therefore$ g attains an absolute minimum value at $x = 0$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.